

Entropy-Regularized, Arbitrage-Constrained Reconstruction of European Option Price Curves

A measure-theoretic formulation, literature review, contradiction audit, and finite-grid estimator

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Abstract

This paper studies reconstruction of an incomplete cross-section of European option prices at a fixed maturity. The primitive object is a probability measure for the forward-normalized terminal underlying, not an implied-volatility curve and not necessarily a smooth density. Calls and puts are linear observations of this common measure. Nonnegativity and the martingale moment condition imply put–call parity, monotonicity, convexity, sharp endpoint behavior, and the usual price bounds. On a finite state grid, the measure is estimated by a convex program combining relative entropy, an optional grid-scaled roughness penalty, bid–ask interval constraints, and penalized quote relaxation. The formulation is corrected to distinguish continuous theory from its atomic numerical approximation, to state the assumptions under which forward normalization is valid, and to avoid claiming cross-maturity arbitrage freedom from independent slice fits.

The paper adds a focused literature review, characterization and uniqueness results, feasibility conditions, identification bounds, proof sketches, a reproducible numerical illustration, and a contradiction audit. Several tempting claims are rejected: finite option quotes do not identify a unique terminal distribution; a truncated grid is not innocuous; smoothness is not a no-arbitrage condition; independent maturity fits need not admit a common martingale; and direct implied-volatility methods are not intrinsically invalid. The resulting method is best understood as a transparent, fixed-maturity, arbitrage-constrained selection rule whose uncertainty must be reported rather than hidden.

Keywords: option interpolation; risk-neutral measure; maximum entropy; static arbitrage; inverse problems; state prices; convex optimization; identification.

1 Problem, scope, and corrected contribution

An option exchange reveals only finitely many bid and ask quotations. A complete price curve must therefore be reconstructed from partial and noisy information. The relevant question is not merely how to fill an empty cell, but whether all filled and observed prices can coexist under a positive linear pricing rule.

This paper addresses one maturity at one observation time. Its scope is deliberately limited to European options on a nonnegative terminal underlying, with a common payoff convention and settlement date. The main contribution is a synthesis rather than a claim of a new asset-pricing theorem: state-price recovery, entropy selection, shape-constrained estimation, and interval-valued observations are assembled into one auditable convex program. The revisions make five distinctions explicit.

1. A probability *measure* is primitive; a density exists only under additional absolute-continuity assumptions.
2. Static no-arbitrage in strike is a theoretical property of the generated call curve. Smoothness is a numerical preference, not an arbitrage axiom.
3. A finite grid produces an atomic distribution and a piecewise-linear call curve. It approximates, but is not identical to, a smooth continuous density model.
4. Entropy regularization selects one admissible measure relative to a stated prior. It does not reveal “the” uniquely identified market distribution from finitely many quotes.
5. Independent fixed-maturity fits do not establish cross-maturity or dynamic arbitrage freedom.

The appropriate conclusion is therefore conditional. This architecture is attractive when transparent missing-strike recovery, interval consistency, and a common terminal law are more important than a low-dimensional smile formula. It does not dominate every parametric or implied-volatility method in every operational setting.

2 Financial assumptions and normalization

Fix observation time t and maturity $T > t$. The analysis uses the following assumptions.

A1. Contract class.

Calls and puts are European, have the same strike, maturity, underlying, exercise convention, and settlement rule. Early exercise, discrete settlement mismatches, and American-option features are excluded.

A2. Pricing representation.

There is a positive discount factor $D_{t,T}$, a strictly positive forward $F_{t,T}$, and a pricing measure under which

$$C(K, T) = D_{t,T} \mathbb{E}[(S_T - K)^+], \quad P(K, T) = D_{t,T} \mathbb{E}[(K - S_T)^+],$$

with $\mathbb{E}[S_T] = F_{t,T}$. This holds directly with deterministic carry under the usual risk-neutral representation, or after a consistent change of numeraire. With stochastic rates or nonstandard collateral, the measure and forward must be derived rather than assumed.

A3. Support.

$S_T \geq 0$ almost surely. Thus the present formulation covers equities, equity indices, and other nonnegative deliverables. A model for rates, spreads, or commodities capable of negative settlement requires a grid on $\mathbb{R}$ and modified endpoint conditions.

A4. Market data.

Each retained quote has a valid bid not exceeding its ask, and timestamps close enough that a common cross-section is meaningful. Quote intervals are evidence, not exact latent values.

Define

$$X_T = \frac{S_T}{F_{t,T}}, \quad k = \frac{K}{F_{t,T}}, \quad c(k) = \frac{C(K, T)}{D_{t,T} F_{t,T}}, \quad p(k) = \frac{P(K, T)}{D_{t,T} F_{t,T}}.$$

Let μ be the law of X_T . Then

$$\mu \geq 0, \quad \mu([0, \infty)) = 1, \quad \int_0^\infty x \mu(dx) = 1. \quad (1)$$

The normalized prices are

$$c(k) = \int_0^\infty (x - k)^+ \mu(dx), \quad p(k) = \int_0^\infty (k - x)^+ \mu(dx). \quad (2)$$

Proposition 1 (Normalized put–call parity). *Under A1–A3, $c(k) - p(k) = 1 - k$ for every $k \geq 0$.*

Proof. The pointwise identity $(x - k)^+ - (k - x)^+ = x - k$, integrated against μ , gives $\int x d\mu - k \int d\mu = 1 - k$. $\square$

Parity is exact only when the financial inputs are aligned. A visible call–put discrepancy can reflect stale quotes, different settlement conventions, an incorrect forward, or executable transaction costs; it is not automatically evidence that one quoted option is “wrong.”

3 Characterization of an admissible call curve

The measure representation gives more than monotonicity and convexity.

Theorem 2 (Fixed-maturity call-curve characterization). *Let $c : [0, \infty) \rightarrow \mathbb{R}$. The following are equivalent.*

- (i) *There exists a probability measure μ on $[0, \infty)$ with unit first moment such that $c(k) = \int (x - k)^+ \mu(dx)$.*
- (ii) *The function c is convex and nonincreasing, $c(0) = 1$, $\lim_{k \rightarrow \infty} c(k) = 0$, and its right derivative at zero satisfies $c'_+(0) \geq -1$.*

For $k \geq 0$, such a curve obeys

$$(1 - k)^+ \leq c(k) \leq 1, \quad -1 \leq c'_+(k) \leq c'_-(k) \leq 0. \quad (3)$$

The representing measure is unique and is recovered distributionally by

$$\mu((k, \infty)) = -c'_+(k), \quad \mu = -dc'_+, \quad (4)$$

with an atom at zero of size $1 + c'_+(0)$. If c is twice differentiable, then $c''(k)$ is the density of the absolutely continuous part.

Proof sketch. The forward direction follows from dominated one-sided differentiation of (2), convexity of each payoff $(x - k)^+$ in k , and monotone convergence at infinity. For the converse, convexity makes c'_+ nondecreasing and right-continuous. Define the tail of a measure on $(0, \infty)$ by $\mu((k, \infty)) = -c'_+(k)$ and put the remaining mass at zero. Integration of this tail yields $c(k) = \int_k^\infty \mu((u, \infty)) du$, while $c(0) = 1$ supplies the unit first moment. The tail function determines μ uniquely. $\square$

Remark 3 (Correction to the density language). The Breeden–Litzenberger relation is naturally distributional. Atoms in μ create kinks in c ; a classical second derivative need not exist. Calling c'' a smooth density without stating regularity excludes economically admissible laws and incorrectly describes a finite-grid solution.

4 Literature review and position of the method

Breeden and Litzenberger established the connection between strike curvature and state-contingent prices, providing the theoretical basis for recovering a pricing distribution from option prices [1]. Rubinstein’s implied binomial tree and related local-volatility constructions pursued a dynamically interpretable tree calibrated to option prices [2]; that objective is richer than the present single-maturity marginal reconstruction and therefore requires additional structure.

Entropy methods address the underdetermination created by finitely many prices. Buchen and Kelly derived maximum-entropy asset distributions consistent with option constraints [3]. Stutzer proposed an entropy-based nonparametric derivative valuation method [4]. Avellaneda and coauthors used relative entropy to reweight simulated paths while matching benchmark instruments [5]. Halperin treated noisy prices through Bayesian entropic inverse theory and emphasized the instability of exact minimum-cross-entropy calibration under noise [6]. Cont and Tankov used relative entropy to regularize ill-posed calibration of jump-diffusion models [7]. These works justify entropy as a transparent selection or regularization device, but none makes the selected posterior independent of its reference measure.

A separate literature imposes shape constraints directly. Aït-Sahalia and Duarte developed nonparametric option-price estimation under monotonicity and convexity constraints [8]. Bondarenko’s positive convolution approximation estimates risk-neutral densities within a positive, regular class and controls overfitting [9]. Davis and Hobson gave conditions under which finite sets of traded option prices are compatible with an arbitrage-free model, clarifying the difference between recognizable misspecification and directly exploitable arbitrage [10]. Fengler constructed arbitrage-free smoothing splines for implied-volatility surfaces through constraints on the associated call-price surface [11].

Finally, parametric volatility methods cannot be dismissed categorically. Gatheral and Jacquier exhibit SVI parameter restrictions sufficient to exclude static arbitrage across strikes and maturities [12]. Their work is a direct counterexample to the claim that interpolation in volatility coordinates is necessarily incoherent. The present method occupies a different niche: it estimates a terminal marginal directly, accommodates missing strikes and interval quotes, and exposes prior and identification sensitivity. Its comparative advantage is interpretability of the state-price object, not universal superiority.

5 From call and put quotes to common intervals

At observed strike k_i , a call quote gives a normalized interval $[b_i^C, a_i^C]$. A put quote gives the call-equivalent interval

$$[b_i^P + 1 - k_i, a_i^P + 1 - k_i]. \quad (5)$$

If both are contemporaneous and refer to identical contracts, both intervals constrain the same latent $c(k_i)$. They should not automatically be intersected and discarded when the intersection is empty: an empty intersection is a diagnostic of asynchronous or inconsistent inputs and can be handled with explicit relaxation.

The bid–ask interpretation requires care. Existence of a coherent shadow price inside every interval is a useful sufficient consistency test for the reconstructed frictionless curve. It is not a complete theorem for a market with general transaction costs, funding asymmetries, position limits, and discrete execution. Accordingly, fitted slack is best labeled a *cross-sectional inconsistency score*, not a guaranteed arbitrage profit.

6 Finite-grid estimator

Choose a state grid

$$0 \leq x_1 < x_2 < \cdots < x_M$$

and let π_j be the probability mass at x_j . Using masses rather than values called “densities” removes quadrature ambiguity. Define

$$A_{ij} = (x_j - k_i)^+, \quad \hat{c}_i = (A\pi)_i. \quad (6)$$

The martingale simplex is

$$\mathcal{P}_M = \left\{ \pi \in \mathbb{R}^M : \pi \geq 0, \mathbf{1}^\top \pi = 1, x^\top \pi = 1 \right\}. \quad (7)$$

Let $\pi_j^0 > 0$ be a reference probability vector. Relative entropy is

$$D_{\text{KL}}(\pi \| \pi^0) = \sum_{j=1}^M \pi_j \log \frac{\pi_j}{\pi_j^0}, \quad 0 \log 0 := 0. \quad (8)$$

On a uniform grid of spacing h , an optional roughness operator may act on the implied grid density $f_j = \pi_j/h$. For interior nodes define

$$(Lf)_j = \frac{f_{j+1} - 2f_j + f_{j-1}}{h^2}. \quad (9)$$

The powers of h matter: an unscaled second-difference penalty changes meaning when the grid is refined. On a nonuniform grid, a consistent finite-volume or spline operator should replace (9).

For quote interval $[b_i, a_i]$, introduce lower and upper violations $v_i^-, v_i^+ \geq 0$:

$$b_i - v_i^- \leq (A\pi)_i \leq a_i + v_i^+. \quad (10)$$

Let $d_i = \max(a_i - b_i, d_{\min})$ set a scale and let $\rho_i > 0$ encode reliability information not already represented by the spread, such as age or depth. The proposed estimator is

$$\begin{aligned} \min_{\pi, v^-, v^+} \quad & \lambda_E D_{\text{KL}}(\pi \| \pi^0) + \lambda_S \|L(\pi/h)\|_2^2 + \lambda_Q \sum_{i=1}^N \rho_i \frac{v_i^- + v_i^+}{d_i} \\ \text{subject to} \quad & \pi \in \mathcal{P}_M, \quad v^- \geq 0, \quad v^+ \geq 0, \quad \text{and (10)}. \end{aligned} \quad (11)$$

This scaling corrects a weakness of weighting an L^1 slack directly by $1/(s_i^2 + \varepsilon)$: that expression mixes units and may count the spread twice. Other convex losses, including quadratic or Huber penalties on violations, are possible, but they change robustness and should be declared.

Lemma 4 (Grid feasibility). *The martingale simplex $\mathcal{P}_M$ is nonempty if and only if $1 \in [x_1, x_M]$. If 1 is an endpoint, every feasible distribution may be forced toward a degenerate boundary configuration; in practice the grid should strictly straddle one.*

Proof. The mean $x^\top \pi$ is a convex combination of the grid points, so it belongs to their convex hull $[x_1, x_M]$. Conversely, if one lies in this interval, a mixture of two bracketing nodes has mean one. $\square$

Proposition 5 (Existence, convexity, and uniqueness). *Assume $1 \in [x_1, x_M]$, $\pi_j^0 > 0$, and $\lambda_E > 0$. Then (11) is feasible and has a solution. Its fitted mass vector $\hat{\pi}$ is unique. At the optimum, the slack variables are the unique componentwise minimal violations induced by $\hat{\pi}$.*

Proof. The quote constraints can always be satisfied by sufficiently large nonnegative slack once $\mathcal{P}_M$ is nonempty. Minimization may be reduced to the compact set $\mathcal{P}_M$ by substituting the minimal hinge violations. The resulting objective is lower semicontinuous and hence attains its minimum. Relative entropy is strictly convex in π when the reference is strictly positive; adding convex terms preserves strict convexity, giving a unique mass vector. Positive redundant slack would strictly increase the objective. $\square$

If $\lambda_E = 0$, uniqueness may fail even when a roughness penalty is present, because the penalty can have a nontrivial nullspace on the feasible affine set. Convexity alone never implies uniqueness.

7 What the estimator guarantees, and what it does not

For every $\hat{\pi} \in \mathcal{P}_M$, define

$$\hat{c}(k) = \sum_{j=1}^M (x_j - k)^+ \hat{\pi}_j, \text{quad}\hat{p}(k) = \sum_{j=1}^M (k - x_j)^+ \hat{\pi}_j. \quad (12)$$

These curves satisfy parity, the bounds in (3), monotonicity, and convexity for *all* $k \geq 0$, not only at quoted strikes. The result follows from the payoff representation, so separate finite-difference shape constraints on generated grid prices are redundant.

However, the finite-grid curve is piecewise linear with kinks at the state nodes. Its second distributional derivative is the atomic measure $\sum_j \hat{\pi}_j \delta_{x_j}$. Interpreting the node masses divided by h as a smooth density is a visualization convention or a further model, not a theorem. The roughness penalty modifies the selected atomic approximation but is not needed for absence of butterfly arbitrage.

The upper grid boundary also has substantive meaning. For every $k \geq x_M$, (12) gives $\hat{c}(k) = 0$; the model asserts bounded terminal support. A narrow grid can therefore manufacture tail conclusions. Production use requires boundary-mass diagnostics, enlargement tests, or explicit tail buckets/extrapolation. The same concern applies near zero.

8 Identification and uncertainty

Finite option observations impose finitely many linear functionals on a distribution. Unless the state space is deliberately very small, they cannot identify every mass.

Proposition 6 (Generic finite-grid nonidentification). *Let*

$$B = \begin{bmatrix} \mathbf{1}^\top \\ x^\top \\ A \end{bmatrix}.$$

If $\text{rank}(B) < M$ and there exists a strictly positive feasible π , then exact price constraints generically admit infinitely many feasible distributions in a neighborhood of π .

Proof. Choose nonzero $z \in \ker B$. For sufficiently small $|\varepsilon|$, positivity of π implies $\pi + \varepsilon z \geq 0$, while $B(\pi + \varepsilon z) = B\pi$. Thus mass, mean, and all quoted prices remain unchanged along a nontrivial line segment. $\square$

Entropy converts this set into a unique selected estimate; it does not eliminate epistemic uncertainty. For any new payoff vector g , honest identification bounds are obtained from the linear programs

$$\underline{V}_g = \min_{\pi \in \mathcal{F}} g^\top \pi, \text{quad} \bar{V}_g = \max_{\pi \in \mathcal{F}} g^\top \pi, \quad (13)$$

where $\mathcal{F}$ includes the mass, mean, and unrelaxed quote-interval constraints. If slack is necessary, the uncertainty set should specify an allowed slack budget rather than silently using the point estimator's violations. Bounds, prior sensitivity, and bootstrap or perturbation analysis answer different questions and should not be conflated.

9 Contradiction audit and rejected alternatives

Table 1 records the strongest objections to the method and the corrected conclusions. “Rejected” means rejected for the stated reconstruction objective, not mathematically impossible in all settings.

Table 1: Contradictions, failure mechanisms, and corrected conclusions.

Claim or alternative	Why the unqualified claim fails	Correct conclusion
“The market determines one density.”	Finitely many prices leave a nullspace of distributions with identical observed values.	Entropy selects one prior-relative member. Report linear-program bounds and sensitivity.
“Copying put volatility to the call is wrong.”	Under the same forward, discount factor, maturity, and model, parity-consistent European calls and puts have the same Black implied volatility.	Copying is a valid local representation when inputs are aligned. It cannot recover a strike where both sides are absent and does not impose global strike convexity.
“Direct volatility interpolation is inherently arbitrageable.”	Arbitrage-free SVI/SSVI and constrained spline constructions are counterexamples.	Unconstrained interpolation can fail; constrained volatility methods remain legitimate competitors.
“A smooth density is guaranteed.”	A finite grid produces atoms, while arbitrary admissible laws may contain atoms.	Smoothness is optional regularization. Use distributional derivatives and distinguish masses from densities.
“Exact quote fitting is more faithful.”	Asynchronous or crossed information can make the interval system infeasible; exact fitting amplifies noise.	Use interval constraints, explicit slack, quote-age controls, and diagnostics.
“The grid is merely numerical.”	Truncation forces zero call values above x_M and can distort tail mass and moments.	Treat the domain as an assumption; test enlarged grids and boundary mass.

Claim or alternative	Why the unqualified claim fails	Correct conclusion
“A second-difference penalty is objective.”	Its magnitude depends on coordinate, spacing, boundary convention, and whether it acts on masses or densities.	Scale the operator, document boundary conditions, and perform grid-refinement checks.
“Independent maturity slices form a surface.”	Equal means alone do not imply a martingale coupling; later marginals must dominate earlier ones in convex order under a common numeraire.	Fixed-maturity validity is local. Joint fitting needs explicit calendar/convex-order constraints.
“Large λ_Q restores truth.”	It only forces proximity to possibly stale or erroneous quotes and can transmit data defects into the distribution.	Select penalties by held-out reconstruction and stability, and retain slack as evidence of conflict.
“The prior disappears with enough data.”	Observed strikes may be dense in the center yet leave tails weakly identified.	Diagnose prior sensitivity by region and payoff; never infer unobserved tail moments without bounds.
“Black implied volatility always exists as a finite number.”	At intrinsic value the limiting volatility is zero; at the upper price bound it can diverge, and numerical inversion becomes ill-conditioned near either boundary.	Invert only admissible interior prices, define boundary conventions, and retain prices as the authoritative output.

10 Cross-maturity consistency

Suppose maturities $T_1 < T_2$ are expressed under a common normalization in which the relevant state process is a martingale and both marginals have mean one. A necessary condition for a martingale coupling is convex order:

$$\mu_{T_1} \preceq_{cx} \mu_{T_2} \iff \int \varphi d\mu_{T_1} \leq \int \varphi d\mu_{T_2} \text{ for every convex integrable } \varphi. \quad (14)$$

For call payoffs, this implies $c(k, T_1) \leq c(k, T_2)$ at every normalized strike in that common convention. Under standard conditions, convex order is also sufficient for existence of a martingale coupling, but it does not specify a unique dynamic process.

Independent slice fitting can violate (14). For example, let the earlier marginal put probability 1/2 at 0.5 and 1/2 at 1.5, and let the later marginal be a point mass at 1. Both have mean one, yet the earlier call at $k = 1$ costs 0.25 while the later call costs zero. A martingale cannot lose conditional dispersion in this way. Smoothness in maturity or a Wasserstein penalty does not repair the contradiction; the ordering constraint itself must be imposed.

With stochastic rates, dividends, collateral, or changing numeraires, even the comparison in (14) requires care. A production multi-maturity model must specify the tradable numeraire and

map quoted monetary prices into a common martingale framework. This paper therefore claims fixed-maturity strike consistency only.

11 Reproducible numerical illustration

Consider $F_{t,T} = 100$, $D_{t,T} = 0.99$, and normalized call-equivalent intervals

k	0.90	1.00	1.10
$[b, a]$	$[0.115, 0.120]$	$[0.055, 0.060]$	$[0.022, 0.026]$

Use 47 equally spaced state nodes from 0.20 to 2.50. Let the reference masses be obtained by integrating a lognormal law with log-volatility 0.25 and exponentially tilting the discrete masses so that their mean is exactly one. Set $\lambda_S = 0$ and first solve the no-slack minimum-relative-entropy problem. A numerical solution to tolerance 10^{-10} produces the fitted normalized calls

$$(\hat{c}(0.90), \hat{c}(1.00), \hat{c}(1.10)) = (0.1200, 0.0600, 0.0260)$$

and the missing-strike estimate

$$\hat{c}(1.05) = 0.04015.$$

The corresponding monetary call is $DF\hat{c} = 99(0.04015) \approx 3.975$. Parity gives the normalized put $\hat{p}(1.05) = 0.09015$ and monetary put approximately 8.925.

The point estimate is stable under several modest prior/grid changes, but this fact must not be mistaken for identification. On the same 47-node grid, linear programming over *all* probability vectors satisfying the three quote intervals and the two moment constraints yields

$$0.02279 \leq c(1.05) \leq 0.04300,$$

while the admissible probability above 1.5 ranges from 0 to 0.05778 and the second moment ranges from 1.0166 to 1.0670. The central missing price is substantially constrained by neighboring calls, but the tail is not. This is precisely the distinction that a single smooth fitted curve can conceal.

For comparison, using a narrower $[0.4, 1.8]$ grid gives $\hat{c}(1.05) = 0.04013$ in the entropy-only fit, but it assigns about 6.95×10^{-5} mass to the two boundary nodes. Boundary mass is small here; in real sparse data it may not be. The diagnostic, not the reassuring outcome of this example, is the transferable lesson.

12 Hyperparameters, diagnostics, and falsification tests

Regularization parameters cannot be justified by aesthetic smoothness alone. A defensible selection protocol separates fit, stability, and identification.

1. **Held-out quote reconstruction.** Remove selected strikes or one side of parity pairs, fit on the remainder, and score whether reconstructions fall within the held-out intervals.
2. **Perturbation stability.** Jitter forwards, discounts, bids, asks, and quote timestamps within plausible error ranges. Report changes in prices and functionals, not only changes in node masses.
3. **Prior sensitivity.** Repeat with several positive references having different central dispersion and tail thickness. A payoff whose value moves materially is prior-dominated.

4. **Grid refinement and expansion.** Refine h , enlarge both boundaries, and rescale L consistently. Check prices, moments, and boundary mass.
5. **Slack diagnostics.** Report normalized violations v_i^-/d_i and v_i^+/d_i , their concentration by quote age and strike, and the solution path as λ_Q varies.
6. **Dual diagnostics.** Convex solvers supply multipliers that identify influential moment and quote constraints. Large multipliers warn that small input changes may materially alter the solution.
7. **Identification intervals.** Compute (13) for every reported missing price, digital probability, and moment of practical interest.

A method is falsified for a requested use when its relevant output is unstable under reasonable inputs, its identification interval is too wide, its solution depends materially on artificial grid boundaries, or required quote slack exceeds a declared tolerance. Convergence of an optimizer is necessary but not evidentiary validation.

13 Algorithmic procedure

For each observation time and maturity:

1. Validate that contracts are European and aligned in settlement and timestamps. Estimate $F_{t,T}$ and $D_{t,T}$ consistently; where possible, compare the forward inferred from parity with the external forward.
2. Normalize strikes and bid–ask prices. Convert puts to call-equivalent intervals using (5). Preserve the raw source, age, spread, and liquidity fields.
3. Run pre-fit interval checks for crossed markets, intrinsic/upper-bound violations, and obvious monotonicity or convexity conflicts. Flag rather than silently delete.
4. Choose a grid that strictly straddles one and extends beyond relevant strikes. Construct a strictly positive reference mass vector and document its origin.
5. Solve (11) with a convex optimization package capable of reporting primal and dual residuals. Verify moment constraints and recompute objective components independently.
6. Generate calls and puts from (12). Classify each output as observed-inside-spread, relaxed, interpolated, or extrapolated.
7. Compute identification bounds and sensitivity diagnostics for material outputs. Reject or qualify outputs that fail predetermined thresholds.
8. Only then invert interior prices to implied volatility. Retain reconstructed prices and provenance as the authoritative stored objects.

14 Limitations

The method remains assumption-laden. It requires a pricing representation, a correctly measured forward and discount factor, and comparable European contracts. The positive-support formulation excludes instruments capable of negative terminal values. The selected distribution depends on the

prior, grid, slack loss, and roughness convention. Sparse quotes weakly identify digital probabilities and tail moments. Bid–ask intervals may be stale, non-executable in size, or contaminated by data errors. A common shadow curve inside spreads is not a complete transaction-cost model.

The fitted law is a pricing measure, not a physical forecast. Risk premia generally separate risk-neutral tail probabilities from real-world frequencies. Nor does a sequence of independently fitted laws define a dynamically arbitrage-free process. Exotic pricing, hedging, and path-dependent risk require a joint dynamic model or robust bounds over admissible couplings.

Finally, the numerical illustration establishes feasibility and exposes identification width; it is not an empirical comparison. Claims of forecasting accuracy, execution quality, or superiority over SVI, splines, or local volatility require out-of-sample data and a predeclared scoring protocol.

15 Conclusion

The mathematically defensible core of density-based option reconstruction is simple. Under explicit pricing and support assumptions, a unit-mean probability measure generates calls and puts as linear payoff integrals. That representation automatically produces parity, strike monotonicity, convexity, bounds, and endpoint behavior. Relative entropy and optional roughness regularization then select a stable finite-grid approximation compatible with interval-valued observations, while explicit slack records conflicts rather than hiding them.

The same first-principles analysis limits the claims. A finite quote set does not identify a unique law; entropy is prior-relative; a grid imposes tail assumptions; smoothness is not arbitrage freedom; call–put volatility transfer is locally valid under parity; constrained implied-volatility methods can also be arbitrage-free; and cross-maturity consistency requires convex-order or stronger dynamic conditions. Subject to these qualifications, the estimator is a coherent fixed-maturity reconstruction layer. Confidence is high in its stated strike-wise guarantees, conditional on A1–A4 and correct numerical implementation; confidence in any particular reconstructed tail quantity must be determined from identification and sensitivity diagnostics, not from the smooth appearance of the fitted curve.

A Additional mathematical notes

A.1 The bounds in (3)

Since $x \mapsto (x - k)^+$ is convex, Jensen’s inequality gives $c(k) \geq (\mathbb{E}[X_T] - k)^+ = (1 - k)^+$. Also $0 \leq (x - k)^+ \leq x$ for $x, k \geq 0$, hence $0 \leq c(k) \leq \mathbb{E}[X_T] = 1$.

A.2 Minimal slack after eliminating auxiliary variables

For fixed π , the unique minimizing violations in (10) are

$$v_i^-(\pi) = (b_i - (A\pi)_i)^+, \quad v_i^+(\pi) = ((A\pi)_i - a_i)^+.$$

Thus (11) is equivalent to minimizing over $\mathcal{P}_M$ a strictly convex entropy term plus a convex quadratic roughness term and weighted hinge losses. This form is useful for theoretical analysis, whereas explicit slacks are often more convenient for conic solvers and reporting.

A.3 Why implied-volatility inversion is secondary

For fixed F, D, K , and maturity, the Black call is strictly increasing in volatility for nondegenerate interior cases. Hence a unique finite implied volatility exists when the reconstructed call lies strictly between the no-arbitrage lower bound and the upper bound. Near either boundary, vega becomes small and inversion is ill-conditioned. Price-space verification must therefore precede numerical inversion.

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